

Math 540

Homework #2

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§17 Homework Problems

Exercise 3. Show that if A is closed in X and B is closed in Y , then $A \times B$ is closed in $X \times Y$.

Proof. We will show that $(A \times B)^c$ is open. First note that:

$$\begin{aligned}(A \times B)^c &= (X \times Y) \setminus (A \times B) \\ &= [(X \setminus A) \times Y] \cup [X \times (Y \setminus B)].\end{aligned}$$

Since $X \setminus A$ is open, surely $(X \setminus A) \times Y$ is open. The same reasoning follows for $X \times (Y \setminus B)$ being open. Thus $(A \times B)^c$ is open, hence $A \times B$ is closed. $\square$

Exercise 6. Let A , B , and A_α denote subsets of a space X . Prove the following.

- (a) If $A \subset B$, then $\overline{A} \subset \overline{B}$.
- (b) $\overline{A \cup B} = \overline{A} \cup \overline{B}$.
- (c) $\overline{\bigcup A_\alpha} \supset \bigcup \overline{A_\alpha}$; give an example where equality fails.

Proof. (a) Let $a \in \overline{A}$. Then for all open $U \subseteq X$ containing a , we have $U \cap A \neq \emptyset$. Since $A \subset B$, surely $U \cap B \neq \emptyset$ as well. Thus $a \in \overline{B}$.

(b) Note that $A \subset A \cup B$ and $B \subset A \cup B$, hence $\overline{A} \subset \overline{A \cup B}$ and $\overline{B} \subset \overline{A \cup B}$. Thus $\overline{A} \cup \overline{B} \subset \overline{A \cup B}$. Conversely, note that $A \cup B \subset \overline{A} \cup \overline{B}$. By definition the closure of a set is the smallest closed set containing it. Since $\overline{A} \cup \overline{B}$ is closed and contains $A \cup B$, it must be the case that $A \cup B \subset \overline{A} \cup \overline{B}$. Thus $\overline{A \cup B} = \overline{A} \cup \overline{B}$.

Let $x \in \bigcup \overline{A_\alpha}$. Then there exists some i such that $x \in \overline{A_i}$. Hence for any open set U containing x , we have $U \cap A_i \neq \emptyset$. It follows that for any open set U containing x , we have $U \cap \bigcup A_\alpha \neq \emptyset$. Thus $x \in \overline{\bigcup A_\alpha}$. A counterexample of equality is as follows: we have $\bigcup_{n=1}^{\infty} (0, \frac{1}{n}) = \overline{\emptyset} = \emptyset$, while $\bigcup_{n=1}^{\infty} \overline{(0, \frac{1}{n})} = \bigcup_{n=1}^{\infty} [0, \frac{1}{n}] = \{0\}$. $\square$

Exercise 11. Show that the product of two Hausdorff spaces is Hausdorff.

Proof. Let X and Y be Hausdorff spaces. Let $(x, y), (x', y') \in X \times Y$. It is enough to assume that these two points differ in one coordinate; i.e., assume without loss of generality that $x = x'$. Since $y, y' \in Y$, we can find open sets $V, V' \subseteq Y$ such that $y \in V$, $y' \in V'$, and $V \cap V' = \emptyset$. Hence $X \times V$ and $X \times V'$ will be disjoint open neighborhoods of (x, y) and (x', y') . Thus $X \times Y$ is Hausdorff. $\square$

Exercise 13. Show that X is Hausdorff if and only if the diagonal $\Delta = \{(x, x) \mid x \in X\}$ is closed in $X \times X$.

Proof. Suppose X is Hausdorff. We will show that Δ^c is open. Let $(x, y) \in \Delta^c$. Then $x \neq y$. Find open sets U containing x and V containing y such that $U \cap V = \emptyset$. Then $(x, y) \in U \times V$, and in particular $(U \times V) \cap \Delta = \emptyset$ (indeed, if we suppose towards contradiction that $(a, b) \in (U \times V) \cap \Delta$, then $a = b$ and $(a, b) \in U \times V$ implies $a \in U \cap V$, a contradiction.) Thus $(x, y) \in U \times V \subseteq \Delta^c$, hence Δ is closed.

Now suppose Δ is closed. Let $x, y \in X$ with $x \neq y$. Then $(x, y) \in \Delta^c$. Since this set is open, we can find a basis element $U \times V$ (where U and V are open) such that $(x, y) \in U \times V \subseteq \Delta^c$. Hence $(U \times V) \cap \Delta = \emptyset$. Now if it were the case that $U \cap V \neq \emptyset$, then $x \in U \cap V$ means $x \in U$ and $x \in V$, which would imply that $(x, x) \in (U \times V) \cap \Delta$, contradiction the conclusion we came to. Thus $U \cap V = \emptyset$, establishing X as Hausdorff. $\square$

Exercise 19. If $A \subset X$, we define the **boundary** of A by the equation $\partial A = \overline{A} \cap \overline{A^c}$.

(a) Show that A° and ∂A are disjoint, and $\overline{A} = A^\circ \cup \partial A$.

(b) Show that $\partial A = \emptyset$ if and only if A is both closed and open.

Proof. (a) We will first prove that $A^\circ \cap \partial A = \emptyset$. Suppose towards contradiction $x \in A^\circ \cap \partial A$. Then $x \in A^\circ$, $x \in \overline{A}$, and $x \in \overline{A^c}$ by definition. If $x \in \overline{A^c}$, then for any open set U containing x , $U \cap A^c \neq \emptyset$. But if $x \in A^\circ$, then we can find an open set V containing x such that $V \subseteq A$. But this means $V \cap A^c = \emptyset$, which contradicts the assumption that $x \in \overline{A^c}$. Thus $A^\circ \cap \partial A = \emptyset$.

Now let $x \in \overline{A}$. We want to show that $x \in A^\circ \cup \partial A$. If $x \in A^\circ$, then we are done. Suppose that $x \notin A^\circ$, we wish to now show that $x \in \overline{A}$ and $x \in \overline{A^c}$. Since we already assumed $x \in \overline{A}$, we only need to show that $x \in \overline{A^c}$. Since $x \notin A^\circ$, for every open set U containing x we have $U \not\subseteq A$. Thus for every open set U containing x we have $U \cap A^c \neq \emptyset$. Thus $x \in \overline{A^c}$, finally establishing $x \in A^\circ \cup \partial A$. Now let $x \in A^\circ \cup \partial A$. If $x \in A^\circ$ then $x \in \overline{A}$. If $x \in \partial A$, then $x \in \overline{A}$ by definition. Thus $\overline{A} = A^\circ \cup \partial A$.

(b) Suppose that $\partial A = \emptyset$. Then $\overline{A} \cap \overline{A^c} = \emptyset$. So for all $x \in X$, $x \notin \overline{A}$ and $x \notin \overline{A^c}$. So there exists an open set U containing x such that $U \cap A = \emptyset$ and there exists an open set V containing x such that $V \cap A^c = \emptyset$. Thus there exists an open set U such that $x \in U \subseteq A^c$ and there exists an open set V such that $x \in V \subseteq A$. Thus A and A^c are open; i.e., A is both open and closed. Conversely, assume A is both open and closed. Suppose towards contradiction $x \in \overline{A} \cap \overline{A^c}$. Since A is both open and closed, $\overline{A} = A$ and $\overline{A^c} = A^c$. Hence $x \in A \cap A^c$. This is a contradiction, hence $\partial A = \emptyset$. $\square$

Exercise 20.

Proof.

$\square$

§18 Homework Problems

Exercise 1. *Prove that for functions $f : \mathbb{R} \rightarrow \mathbb{R}$ the ϵ - δ definition of continuity implies the open set definition.*

Proof. Let $U \subseteq \mathbb{R}$ be open. Let $x_0 \in f^{-1}(U)$. Then $f(x_0) \in U$. Since U is open, find $\epsilon > 0$ such that $B(f(x_0), \epsilon) \subseteq U$. By the ϵ - δ definition of continuity, we can find some $\delta > 0$ such that $x \in B(x_0, \delta)$ implies $f(x) \in B(f(x_0), \epsilon)$. In particular, $f(B(x_0, \delta)) \subseteq B(f(x_0), \epsilon)$. Thus $B(x_0, \delta) \subseteq f^{-1}(B(f(x_0), \epsilon)) \subseteq f^{-1}(U)$. Since x_0 was arbitrary, $f^{-1}(U)$ is open. $\square$

Exercise 2. *Suppose that $f : X \rightarrow Y$ is continuous. If x is a limit point of the subset A of X , is it necessarily true that $f(x)$ is a limit point of $f(A)$?*

Proof. This is not true. As a counter example, consider the constant function $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = 2$. The set of limit points of the set $[0, 1]$ is $[0, 1]$. But the set of limit points of $f([0, 1]) = \{2\}$ is $\emptyset$. $\square$